

Engineering Reboot 2026

Week-02 Engineering Lab Workbook v3.4

LOCK-IN EDITION

Purpose: Self-contained execution workbook. Open this document at 7:00 PM and execute without cross-referencing other manuals. All file locations, prompts, outputs, screenshots, notes, journals, and Git activities are explicitly defined.

Week-02 Directory Structure

Week-02/

Day-06/Exercises, Screenshots, Notes, Journal

Day-07/Exercises, Screenshots, Notes, Journal

Day-08/Exercises, Screenshots, Notes, Journal

Day-09/Exercises, Screenshots, Notes, Journal

Day-10/Projects, Screenshots, Notes, Journal

Day 06 – Conditional Logic

Pre-Flight Checklist

- ☐ VS Code Open
- ☐ Correct Day Folder Exists
- ☐ Screenshots Folder Exists
- ☐ Notes Folder Exists
- ☐ Journal Folder Exists
- ☐ Git Repository Accessible

Required Source Files

Week-02\Day-06\Exercises\age_classifier.py

Week-02\Day-06\Exercises\temperature_advisor.py

Exercise Specification – Age Classifier

Prompt: Input age:

Test Inputs: 5, 13, 35, 70

Expected Outputs: Child, Teen, Adult, Senior respectively.

Notes File

Week-02\Day-06\Notes\Day-06-Notes.md

Journal File

Week-02\Day-06\Journal\Day-06-Journal.md

Git Command Block

```
git status
git add .
git commit -m "Day 06 Complete"
git push
```

Completion Checklist

- ☐ Exercises/Project Complete
- ☐ Test Cases Executed
- ☐ Screenshots Captured
- ☐ Notes Completed
- ☐ Journal Completed
- ☐ Git Commit Completed
- ☐ Git Push Completed

Day 07 – Boolean Logic

Pre-Flight Checklist

- ☐ VS Code Open
- ☐ Correct Day Folder Exists
- ☐ Screenshots Folder Exists
- ☐ Notes Folder Exists
- ☐ Journal Folder Exists
- ☐ Git Repository Accessible

Required Source Files

Week-02\Day-07\Exercises\eligibility_checker.py

Week-02\Day-07\Exercises\building_access_control.py

Notes File

Week-02\Day-07\Notes\Day-07-Notes.md

Journal File

Week-02\Day-07\Journal\Day-07-Journal.md

Git Command Block

```
git status
git add .
git commit -m "Day 07 Complete"
git push
```

Completion Checklist

- ☐ Exercises/Project Complete
- ☐ Test Cases Executed
- ☐ Screenshots Captured
- ☐ Notes Completed
- ☐ Journal Completed
- ☐ Git Commit Completed
- ☐ Git Push Completed

Day 08 – Loops

Pre-Flight Checklist

- ☐ VS Code Open
- ☐ Correct Day Folder Exists
- ☐ Screenshots Folder Exists
- ☐ Notes Folder Exists
- ☐ Journal Folder Exists
- ☐ Git Repository Accessible

Required Source Files

Week-02\Day-08\Exercises\counting_program.py

Week-02\Day-08\Exercises\even_number_generator.py

Week-02\Day-08\Exercises\sum_calculator.py

Notes File

Week-02\Day-08\Notes\Day-08-Notes.md

Journal File

Week-02\Day-08\Journal\Day-08-Journal.md

Git Command Block

```
git status
git add .
git commit -m "Day 08 Complete"
git push
```

Completion Checklist

- ☐ Exercises/Project Complete
- ☐ Test Cases Executed
- ☐ Screenshots Captured
- ☐ Notes Completed
- ☐ Journal Completed
- ☐ Git Commit Completed
- ☐ Git Push Completed

Day 09 – Loop Practice

Pre-Flight Checklist

- ☐ VS Code Open
- ☐ Correct Day Folder Exists
- ☐ Screenshots Folder Exists
- ☐ Notes Folder Exists
- ☐ Journal Folder Exists
- ☐ Git Repository Accessible

Required Source Files

Week-02\Day-09\Exercises\square_pattern.py

Week-02\Day-09\Exercises\right_triangle_pattern.py

Week-02\Day-09\Exercises\inverted_triangle_pattern.py

Week-02\Day-09\Exercises\multiplication_table.py

Pattern Output Examples

Square Pattern Input 5:

```
*****
*****
*****
*****
*****
```

Right Triangle Input 5:

```
*
**
***
****
*****
```

Inverted Triangle Input 5:

```
*****
****
***
**
*
```

Notes File

Week-02\Day-09\Notes\Day-09-Notes.md

Journal File

Week-02\Day-09\Journal\Day-09-Journal.md

Git Command Block

git status

git add .

git commit -m "Day 09 Complete"

git push

Completion Checklist

☐ Exercises/Project Complete

☐ Test Cases Executed

☐ Screenshots Captured

☐ Notes Completed

☐ Journal Completed

☐ Git Commit Completed

☐ Git Push Completed

Day 10 – Mini Project

Pre-Flight Checklist

- ☐ VS Code Open
- ☐ Correct Day Folder Exists
- ☐ Screenshots Folder Exists
- ☐ Notes Folder Exists
- ☐ Journal Folder Exists
- ☐ Git Repository Accessible

Required Source Files

Week-02\Day-10\Projects\menu_driven_calculator.py

Functional Requirements

- FR-01 Menu Display
- FR-02 Addition
- FR-03 Subtraction
- FR-04 Multiplication
- FR-05 Division
- FR-06 Division By Zero Handling
- FR-07 Exit

Notes File

Week-02\Day-10\Notes\Day-10-Notes.md

Journal File

Week-02\Day-10\Journal\Day-10-Journal.md

Git Command Block

```
git status
git add .
git commit -m "Day 10 Complete"
git push
```

Completion Checklist

- ☐ Exercises/Project Complete
- ☐ Test Cases Executed
- ☐ Screenshots Captured
- ☐ Notes Completed
- ☐ Journal Completed
- ☐ Git Commit Completed
- ☐ Git Push Completed

Week-02 Completion Checklist

- ☐ Day-06 Complete
- ☐ Day-07 Complete
- ☐ Day-08 Complete
- ☐ Day-09 Complete
- ☐ Day-10 Complete
- ☐ Repository Up To Date
- ☐ Ready For Week-03